

Intelligent Support Ticket Prioritization System

TAC 459 Final Presentation



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Industry Overview

The ticket classification industry is a fast-growing part of the broader **AI-driven customer support and service automation market**.

Core function: automatically analyze incoming support tickets, so they can be routed and handled efficiently.

Low-end: keyword/rule-based classifiers

Mid-market: ML classifiers inside helpdesk tools

High-end (our product): LLM-powered systems that

- understand context
- classify tickets
- generate replies

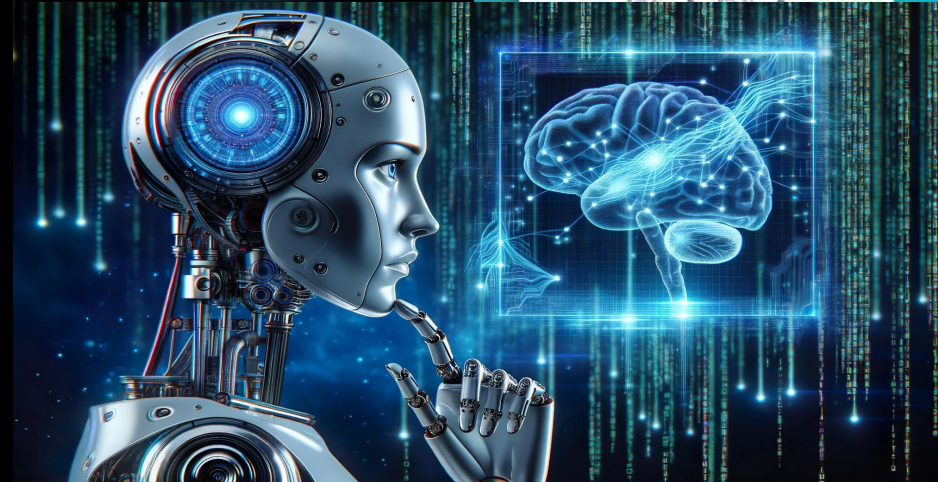
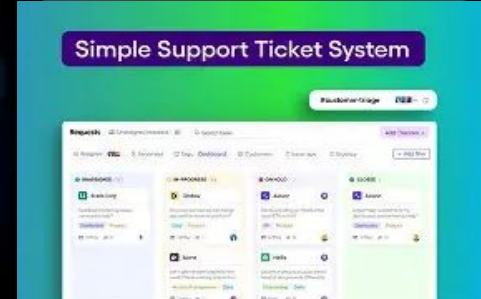
Brief Market Overview

Target Customers:

- SaaS companies, IT service providers, customer support teams, and mid-sized enterprises

Key competitors:

- Zendesk AI, Freshdesk/Freddy AI, ServiceNow, Salesforce Agentforce
- Automated routing/ticket summarization



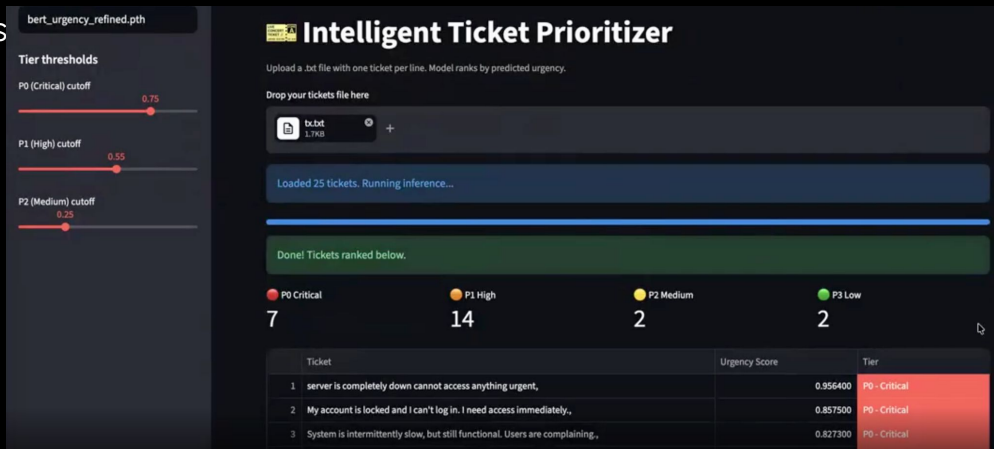
Problem

Support teams today cannot distinguish urgent, high-risk tickets from routine requests.

- All tickets look the same in the queue
- Critical issues get buried and missed
- No scalable way to prioritize urgency

Why this matters:

- B2B SaaS revenue is highly concentrated
- One missed ticket can mean **\$20K–\$100K ARR loss**
- Teams handle **100–500+ tickets/day** → manual triage breaks



[Learn more](#)



Target Users & Market Gap



B2B SaaS has high revenue concentration

One missed ticket = \$20K–\$100K ARR loss

Teams handle 100–500+ tickets/day with limited agents



Mid-market B2B SaaS companies (\$10M–\$100M ARR)

- Primary user: Support team leads and managers responsible for SLA adherence, queue performance, and agent allocation

Why mid-market?

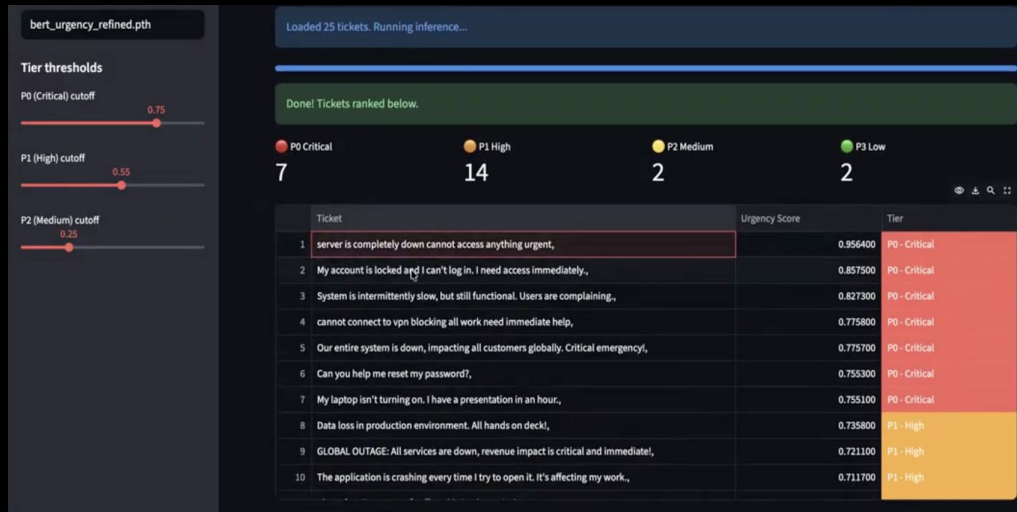
- Startups: too little volume to feel the pain acutely
- Enterprises: build custom solutions in-house
- Mid-market: enough volume to have the problem, not enough resources to solve it

Product Intro

- AI-powered layer for support platforms (Zendesk, Intercom)
- Automatically detects urgency, sentiment, and context using NLP
- Assigns real-time priority scores to every ticket

How It Works

1. Ticket enters system
2. AI analyzes text (sentiment, keywords, context)
3. Priority score assigned
4. Queue reordered + alerts triggered



Features

Core Features

Real-time priority scoring

Queue reordering + alerts

Dashboard for trends & high-risk accounts

LLMs for summary + resolution

Continuous improvement via Memory+RAG

Why It's Different

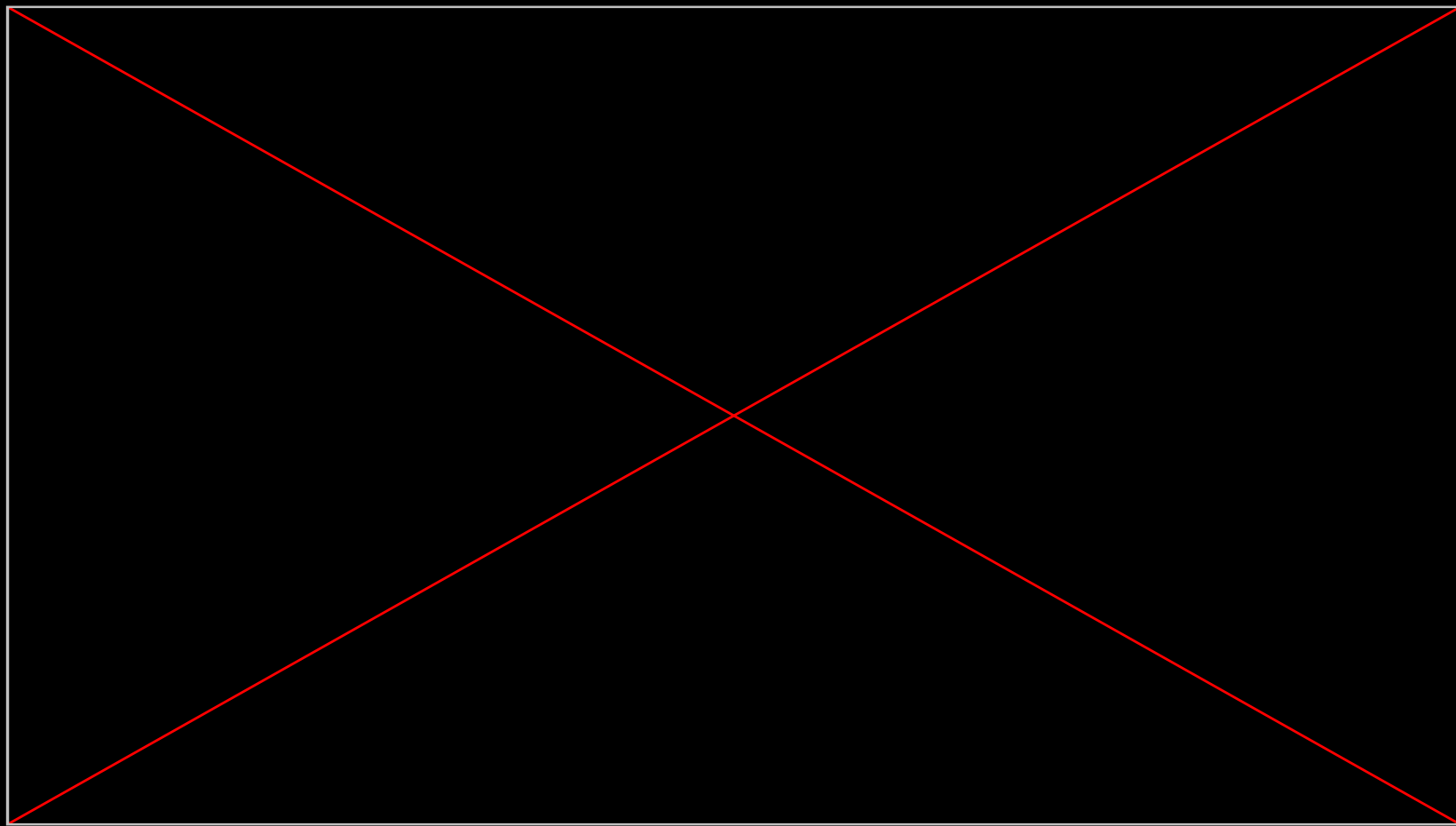
Understands language (not rule-based)

Improves human decision-making, not just automation

Learns company-specific patterns over time

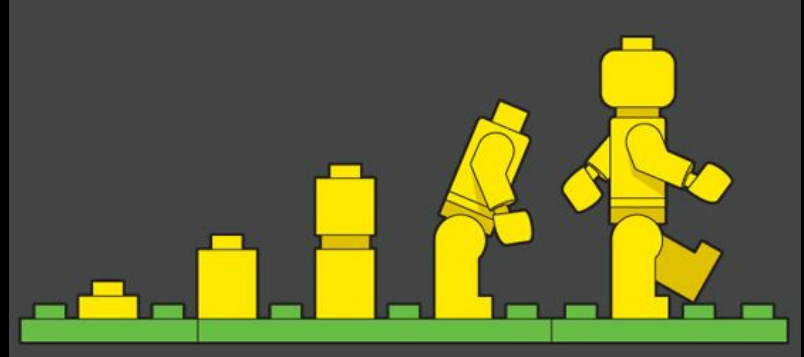
Competitive Landscape

	Rule-Based (Zendesk, Freshdesk, Intercom)	Deflection-Focused (Forethought, ADA)	Enterprise Custom Pipelines	Our Product
NLP / Sentiment	✗	✗	✓	✓
Works on existing stack	✓	✓	✗	✓
Mid-market accessible	✓	✓	✗	✓
Prioritizes human agents	✗	✗	✓	✓



Roadmap Iteration

- Can a fine-tuned BERT predict the *urgency* of tickets?
 - ◆ Thin prioritization layer
- Can a fine-tuned BERT categorize tickets by business impact?
 - ◆ Team-based view
- LLMs lend two helping hands
 - ◆ OpenAI API for ticket summarization and resolution
- RAG + Memory: *cherry on top*
 - ◆ Respond to tickets as you wish, getting better over time



Code + Design

→ Fine-tuned BERT:

- ◆ Fine tune bert-base-uncased on two support ticket NLP datasets (urgency, categories)
- ◆ Warm-up training
- ◆ Saved on + fetched from *HuggingFace*

→ Streamlit service

- ◆ Load models *once* from *HF*; parse tickets; infer
- ◆ Easy dashboard UI

Click a row to select it. Hold Cmd/Ctrl to select multiple for bulk actions.

ID	Summary	Category	+ Urgency Score	Tier
☐	1 I think there might be some unauthorized access on my account. I just got an ...	Access	0.773000	P0 - Critical
☐	2 Our entire production database appears to be down. We started getting alerts ...	Hardware	0.972000	P0 - Critical
☐	3 My desk phone is making a constant buzzing noise when I pick up the receiver...	Hardware	0.755700	P0 - Critical
☐	4 The build pipeline has been failing for the past 2 hours. Every PR is getting...	Internal Project	0.710900	P1 - High
☐	5 We're running into storage limits on our shared team drive. The "Marketing-20...	Storage	0.893600	P1 - High
☒	6 The wifi in conference room B has been spotty all week. Calls drop, screen sh...	Hardware	0.682900	P1 - High
☐	7 Hello, I'd like to formally request a quota increase for our team's Google Dr...	Hardware	0.665200	P1 - High
☐	8 Hi, I'm trying to onboard a new contractor who'll be helping us with the webs...	Hardware	0.643400	P1 - High
☐	9 My laptop won't turn on at all. I tried plugging it into the charger overnight...	Hardware	0.638200	P1 - High
☐	10 The PDF export feature in the reporting tool has been generating corrupt file...	Administrative rights	0.631000	P1 - High

Download ranked CSV

```
# epochs, freeze_mode_name, learning rate
schedule = [
    (3, 'head_only', 2e-5),
    (4, 'top_layers', 1e-5),
    (3, 'all', 5e-6),
]

epoch_num = 0
for (num_epochs, mode, lr) in schedule:

    # branching to actually execute schedule
    if mode == 'head_only':
        # freeze all
        set_bert_layers_frozen(model, frozen=True)
        for param in model.bert.pooler.parameters():
            param.requires_grad = True
    elif mode == 'top_layers':
        # unfreeze top layers
        set_bert_layers_frozen(model, frozen=True)
        set_bert_top_layers_unfrozen(model, layers=[9, 10, 11])
        for param in model.bert.pooler.parameters():
            param.requires_grad = True
    elif mode == 'all':
        # unfreeze all!
        set_bert_layers_frozen(model, frozen=False)

    optimizer = AdamW(
        filter(lambda p: p.requires_grad, model.parameters()),
        lr=lr, weight_decay=0.01
    )
```

```
def predict_urgency(text, model, tokenizer, device, max_len=128):
    text = ' '.join(text.strip().split())
    enc = tokenizer(text, truncation=True, padding='max_length',
                    max_length=max_len, return_tensors='pt')
    with torch.no_grad():
        score = model(enc['input_ids'].to(device), enc['attention_mask'].to(device))
    return round(1.0 - score.item(), 4)
```

Code + Design pt.2

→ LLM Assistance:

- ◆ Summarize current queue (arbitrary urgencies, categories)
- ◆ Draft (and revise) response for specific ticket

→ Memory + RAG

- ◆ Resolved tickets saved to disk
- ◆ RAG employed to draw on resolutions for similar past tickets

```
def embed_texts(texts: List[str]) -> np.ndarray:  
    model = _get_model()  
    return model.encode(texts, convert_to_numpy=True, show_progress_bar=False)
```

 RAG active — pulled 1 similar past ticket(s) from the vector store (top similarity 0.43). Expand below to see what was retrieved.

>  Show 1 retrieved tickets

Edit response before sending:

I understand that the build pipeline has been failing for the past 2 hours, causing integration tests to time out and blocking your team's progress. We are currently investigating the potential network issue between the runners and the test database. I recommend that you keep an eye on the runner logs for any additional insights. We will update you as soon as we have more information or if we require further details from your side.

P1 - High **Hardware** | Urgency: 0.68

The wifi in conference room B has been spotty all week everyone. Other rooms on the same floor seem fine. Co

> 🔍 Why was this classified as Hardware?

 Draft AI Response

```
def draft_ticket_response(ticket_text: str, urgency_tier: str, urgency_score: float, category: str, a  
    client = OpenAI(api_key=api_key)  
    cat_hint = _category_hint(category)  
    cat_line = f"\nCategory guidance: {cat_hint}" if cat_hint else ""  
  
    if urgency_tier.startswith(("P0", "P1")):  
        tone_note = "Tone: decisive and calm. Match the urgency without sounding panicked. No ove  
    else:  
        tone_note = "Tone: warm and efficient. Don't manufacture urgency the ticket doesn't have.  
  
    examples_block = ""  
    if similar_examples: -  
  
    prompt = f""""Draft the body of an email response to this support ticket.{examples_block}-  
  
    resp = client.chat.completions.create(  
        model="gpt-4o-mini",  
        messages=[  
            {"role": "system", "content": _RESPONSE_SYSTEM},  
            {"role": "user", "content": prompt},  
        ],  
        temperature=0.6,  
        max_tokens=350,  
    )  
    return resp.choices[0].message.content.strip()
```

Monetization Strategy

Starter

\$500/month

- Up to 100 tickets/day
- Sentiment + keyword triage
- Queue reordering
- 1 platform integration

Growth

\$1,200/month

- Up to 300 tickets/day
- Full NLP pipeline (BERT + LLM-powered assistant)
- Priority scoring + dashboards
- 2 platform integrations

Enterprise

\$2,500+/month

- Unlimited tickets
- Company-specific fine-tuning
- Predictive churn scoring
- Automated escalations
- Dedicated success manager
- Custom integrations + API

Success Metrics

Operational & Model Performance KPIs

Performance

- Average time to first response for P1 (urgent tickets - target: <2 hrs vs. baseline
- **SLA adherence** - % of urgent tickets responded to within SLA window
- **Queue reorder latency** - how quickly a new high-priority ticket surfaces to the top (target: <30 sec)

Reliability

- System uptime
- **Confidence threshold fallback rate** - % of tickets routed to human review queue (a healthy signal, not a failure metric)
- Pipeline processing time per ticket

Adoption

- Total platforms connected
- % of daily tickets processed through the prioritization engine (**adoption depth**)
- Time-to-value for new customers

Success Metrics

Business, Engagement & Satisfaction Metrics

Engagement

- Alert click-through rate
- Dashboard DAU/WAU among support managers
- Feedback loop participation rate

Retention

- Customer Retention Rate
- Model improvement rate over time per customer
- Expansion Revenue

User Satisfaction

- CSAT scores
- Agent NPS
- Escalation rate reduction

Business Impact

- Churn Reduction
- Retained Revenue from at-risk accounts

Potential Risks & Guardrails

1



Model & AI Risks

- Hallucination/misclassification risk: urgent tickets assigned low priority
- Bias in training data: model trained on Kaggle data
- Keyword gaming: customers learn to use trigger words

2



Privacy & Data Risks

- Sensitive Ticket Content: billing info, personal information
- Customer data isolation: one company's fine-tuning data must never influence another's model

3



Operational Risks

- High-value ticket missed (reputational risk): billing info, personal information
- Integration downtime: API failure with Zendesk/Intercom causes prioritization to stop silently
- Alert fatigue: too many high priority flags devalue signal

Features Improvement

Smarter Feedback Loops

Agents rate whether the AI prioritized correctly –those ratings get batched and used to retrain the model periodically, so it gets smarter the longer you use it.

Multi-Language Support

Expand beyond English – Spanish, French, German support for global SaaS companies using Zendesk.

Trend Analytics

Surface recurring ticket themes before they become escalations – proactive product insight for customer success and product teams.

CRM Integration

Pull in account data from Salesforce or HubSpot so high-value or at-risk customers automatically get higher priority—not just based on what they said, but who they are.

Long-Term Vision

With VC funding, a full ML team, and enterprise compute:

1 The Revenue Intelligence Platform



Expand beyond support tickets –connect every customer signal via sales calls, surveys, and product usage into one score that tells you who's about to leave before they do

2 Acquisition Target



Position our product as a buyout target – companies like Zendesk, Salesforce, or HubSpot would acquire our product to add intelligent triage as an intelligent feature, the same way they've snapped up smaller tools to fill gaps in their platforms

3 The Billion Dollar Path



Capture just 5% of mid-market SaaS companies at our average price point and that's \$180M in revenue. By adding Enterprise customers on top of that the path to \$1B is possible.

Thank you

Ready for what's next?

Let's talk